

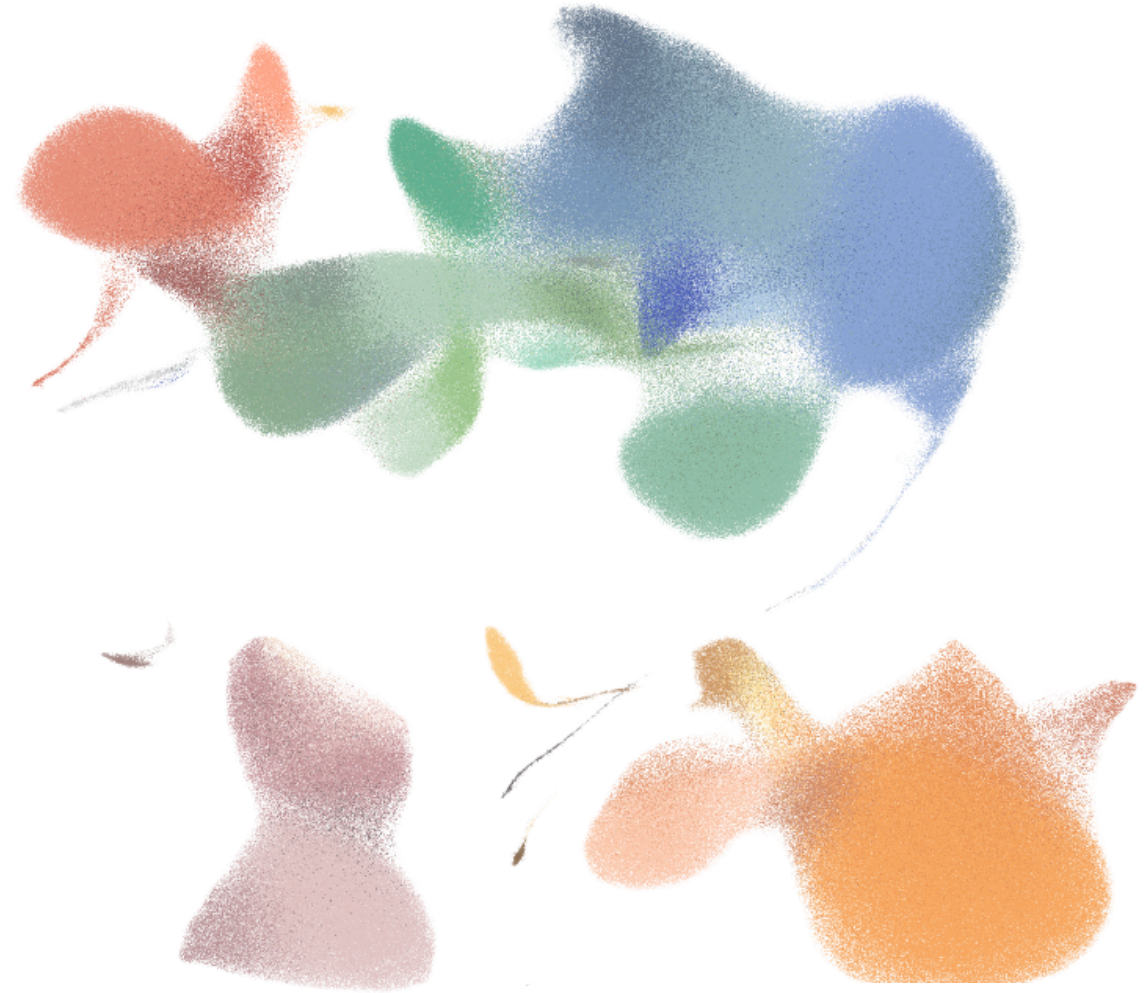
Clustering

COURSE ON SCRNA-SEQ DATA ANALYSIS

Grouping cells by their expression profiles — and answering how many groups there are

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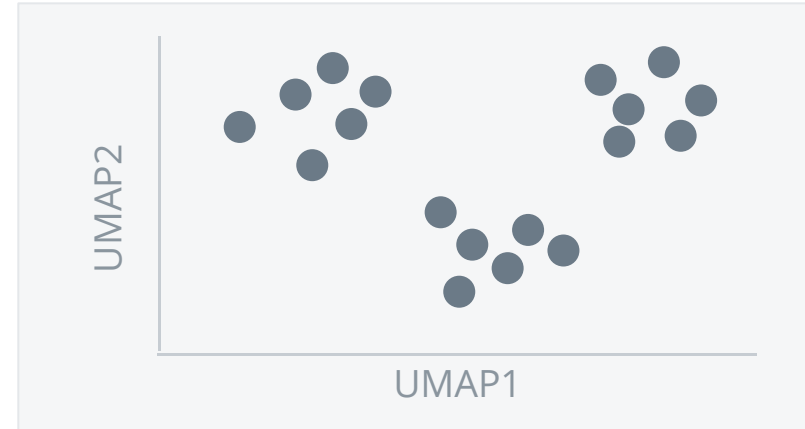
THE GOAL

Grouping cells with similar expression profiles

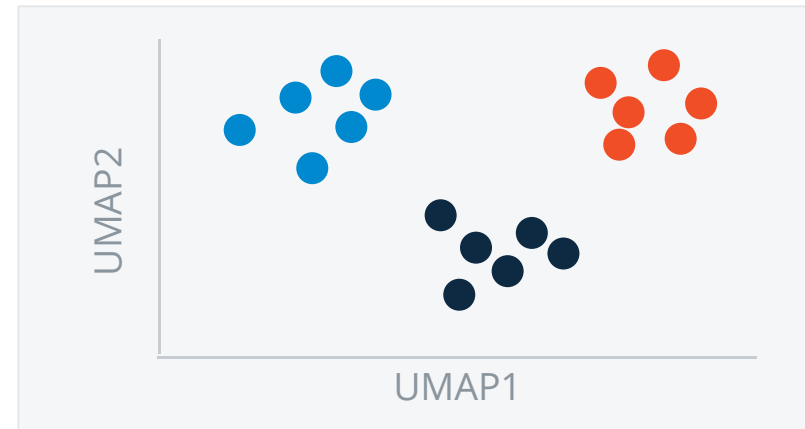
A single-cell dataset is a cloud of tens of thousands of points in a high-dimensional space. Clustering turns that cloud into a small number of named groups we can reason about.

It is an *unsupervised* step: the algorithm is never told what a T cell looks like. It only sees distances between cells.

Everything downstream inherits this grouping — annotation, marker genes, composition tests, trajectories.



unlabelled cells



three clusters

WHERE IT HAPPENS

Clustering runs on the reduced embedding, not on the raw counts

01

QC & filtering

02

Normalisation & HVGs

03

PCA & integration

04

Clustering

05

Markers & annotation

Cluster on this

The first 20–50 principal components, or the integrated latent space. Denoised, and still a faithful description of distance between cells.

Never on this

The two UMAP coordinates. UMAP distorts distances to make a readable picture — it is a display, not a measurement.

PART ONE

How the algorithms work

LANDSCAPE

Three families of clustering are used on single-cell data

***k*-means**

Place k centres, assign each cell to the closest one, move the centres, repeat.

Fast, needs k in advance

Graph-based

Connect each cell to its nearest neighbours, then cut the network into communities.

Default in Seurat & Scanpy

Hierarchical

Merge the closest cells step by step and cut the resulting tree at a chosen height.

Scales poorly to 10^5 cells

ALGORITHM ONE

k-means

K-MEANS CLUSTERING

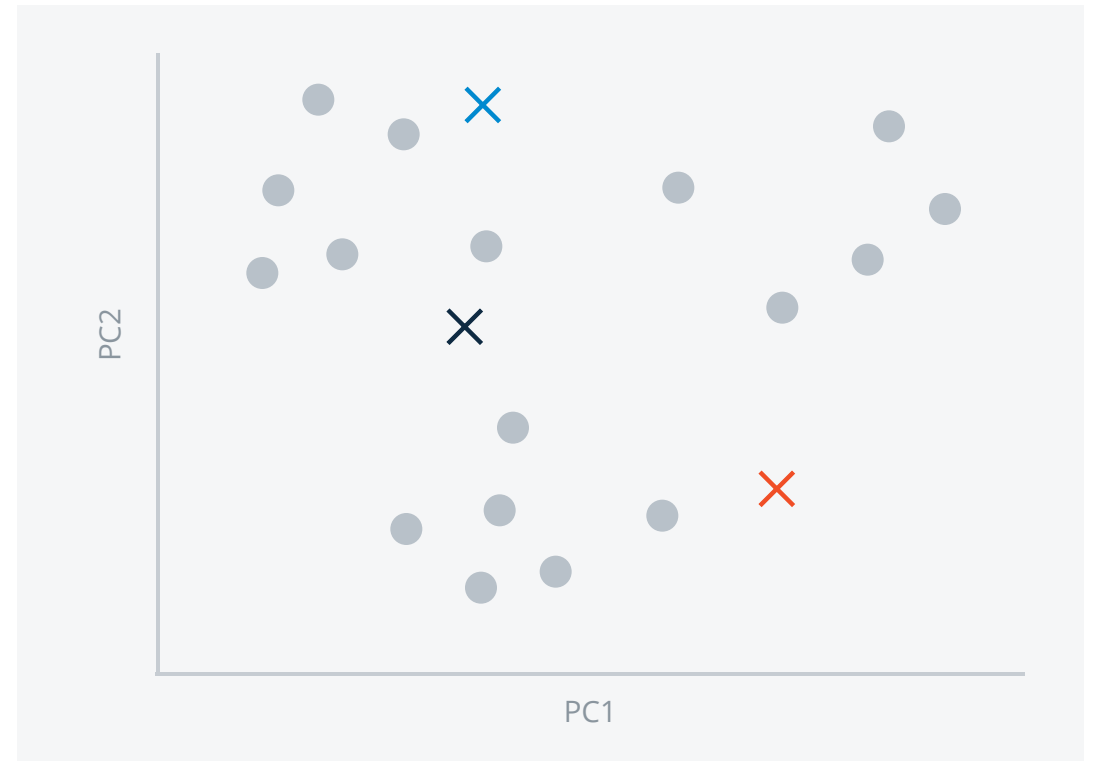
k-means asks you to decide the number of clusters first



K-MEANS CLUSTERING

k-means asks you to decide the number of clusters first

- 1 Choose k and drop k centres at random.
- 2 Assign every cell to its nearest centre.
- 3 Move each centre to the mean of its cells.
- 4 Repeat until no centre moves.



$k = 3$ · centres dropped at random

K-MEANS CLUSTERING

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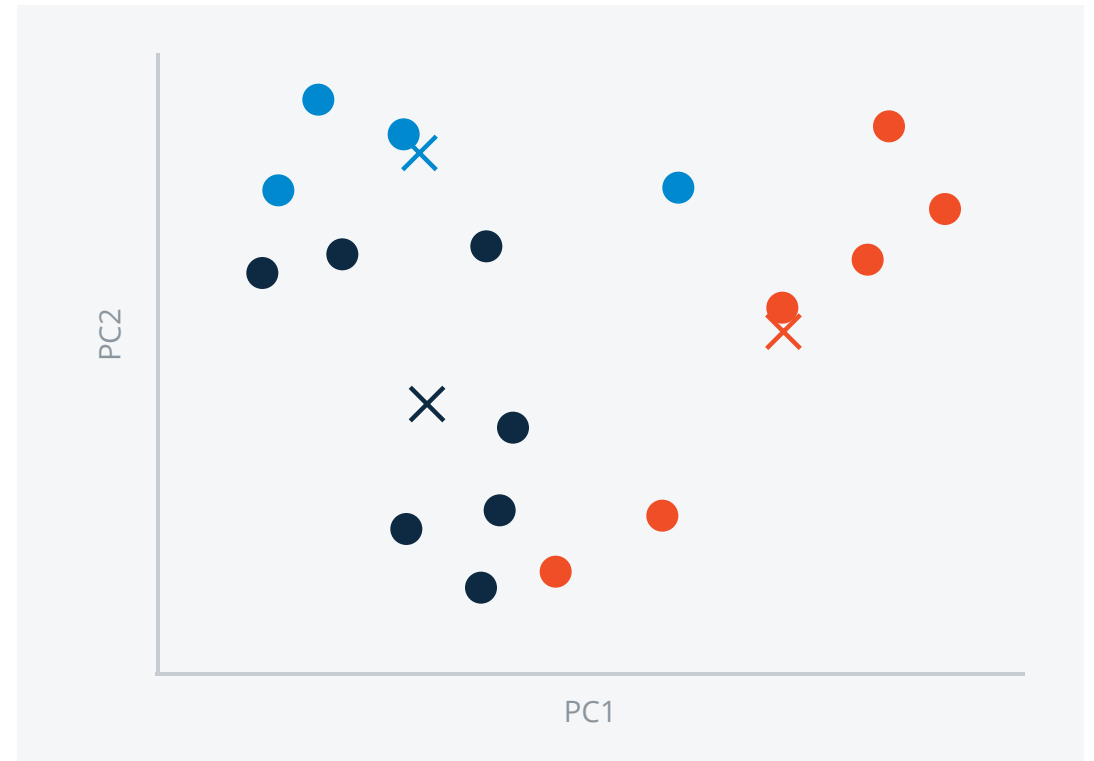


every cell takes its nearest centre · 7 / 4 / 6 cells

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each centre moves to the mean of its own cells

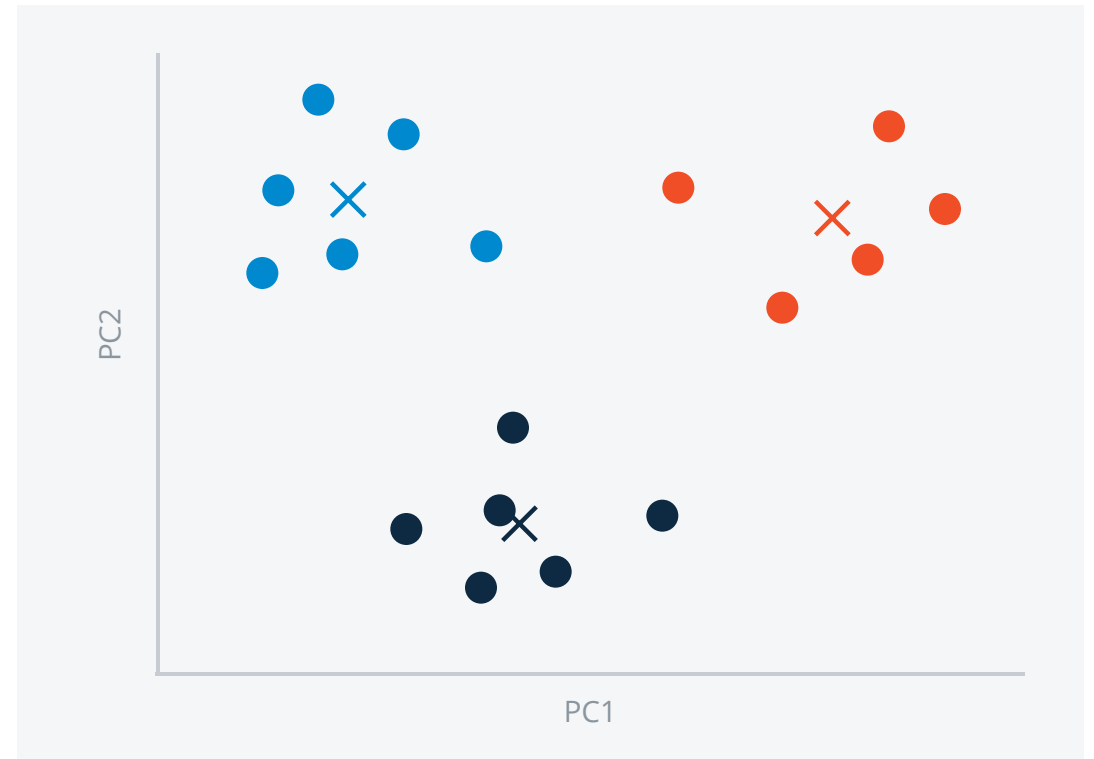
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The starting centers are random, so the same k can converge to different answers.

The catch: it favours round, similarly sized clusters. Rare cell types get absorbed into their large neighbours.



converged after 4 passes · 6 / 6 / 5 cells

ALGORITHM TWO

Graph-based clustering

GRAPH-BASED CLUSTERING

A kNN graph turns distances into neighbourhoods

- 1 **Pick k — the number of neighbours per cell.**
- 2 Link each cell to its k nearest neighbours.
- 3 Weight each link by shared neighbours.
- 4 Cut the weak links — clusters remain.

Real data uses $k = 10\text{--}30$. Small k splits real types apart; large k melts them together.



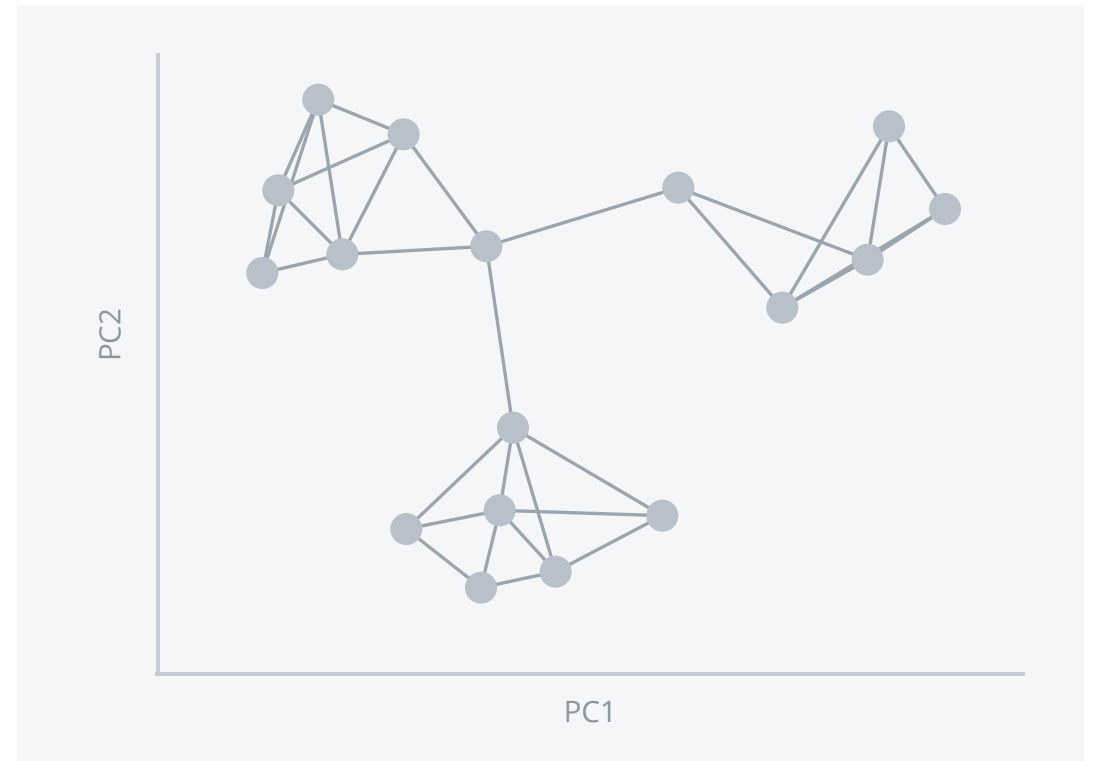
17 cells · $k = 3$ · no links yet

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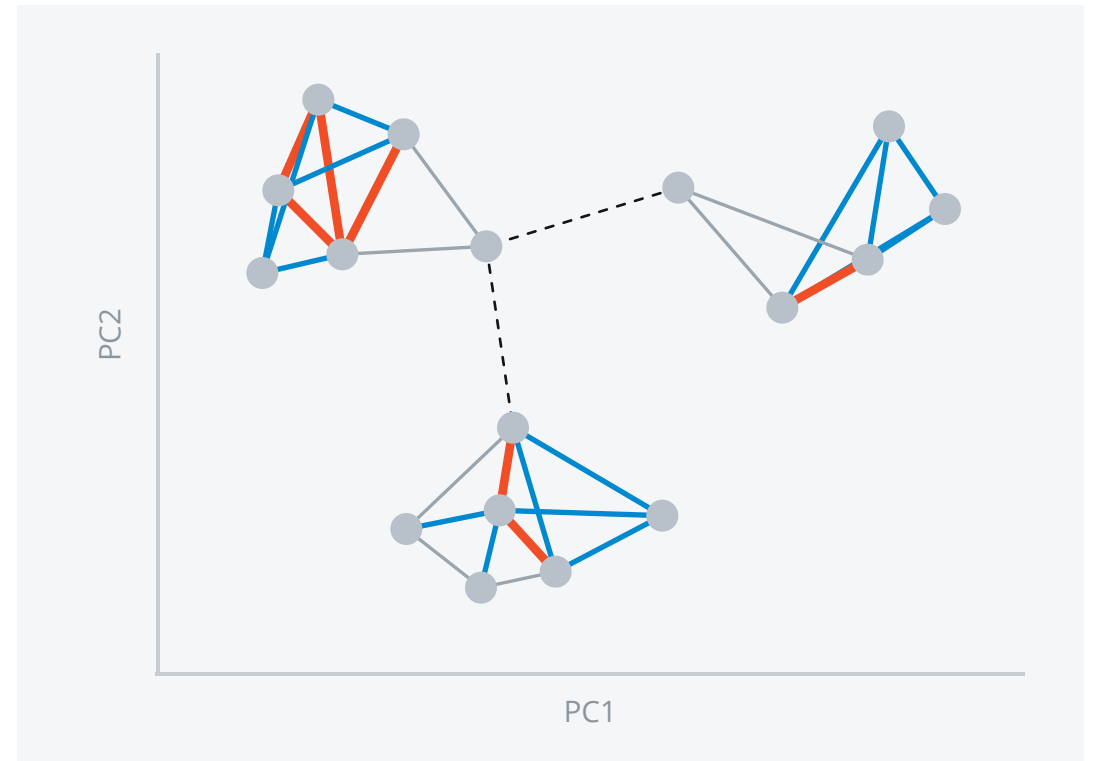
each cell keeps its 3 nearest · 32 links

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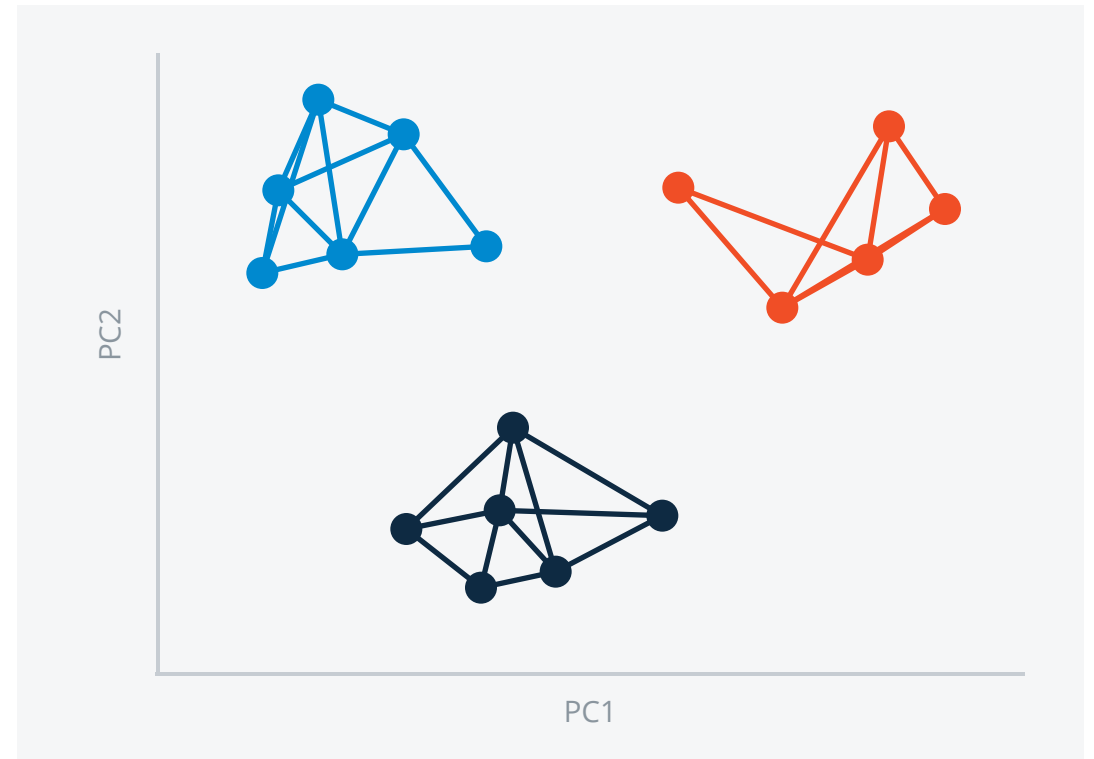
shared: orange 3 · blue 2 · grey 1 · dotted 0

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two zero-shared links cut · three communities

COMMUNITY DETECTION ALGORITHMS FOR GRAPH-BASED CLUSTERING

Louvain and Leiden cut the graph where edges are sparse

Both are *community detection* algorithms. They look for groups of nodes with many edges inside the group and few edges leaving it — formally, they maximise **modularity**.

Louvain

Greedy moves single cells between communities, then collapses each community into one node and repeats.

Can return communities that are internally disconnected.

Leiden

Adds a refinement pass that guarantees every community is connected and converges faster on large graphs.

The recommended default today.

ALGORITHM THREE

Hierarchical clustering

HIERARCHICAL CLUSTERING

Merging the closest pairs builds a tree

- 1 Start with every cell as its own cluster.
- 2 Merge the two closest clusters.
- 3 Repeat until one tree is left.
- 4 Cut the tree where the gap is largest.

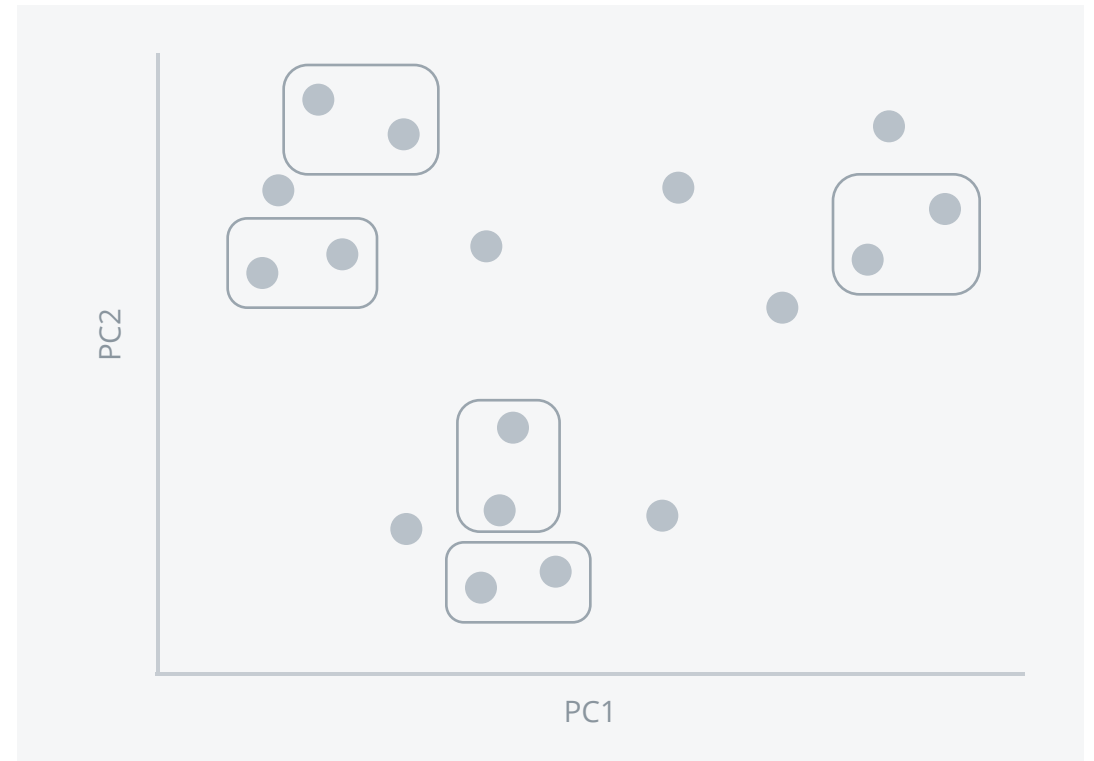


17 cells · 17 clusters

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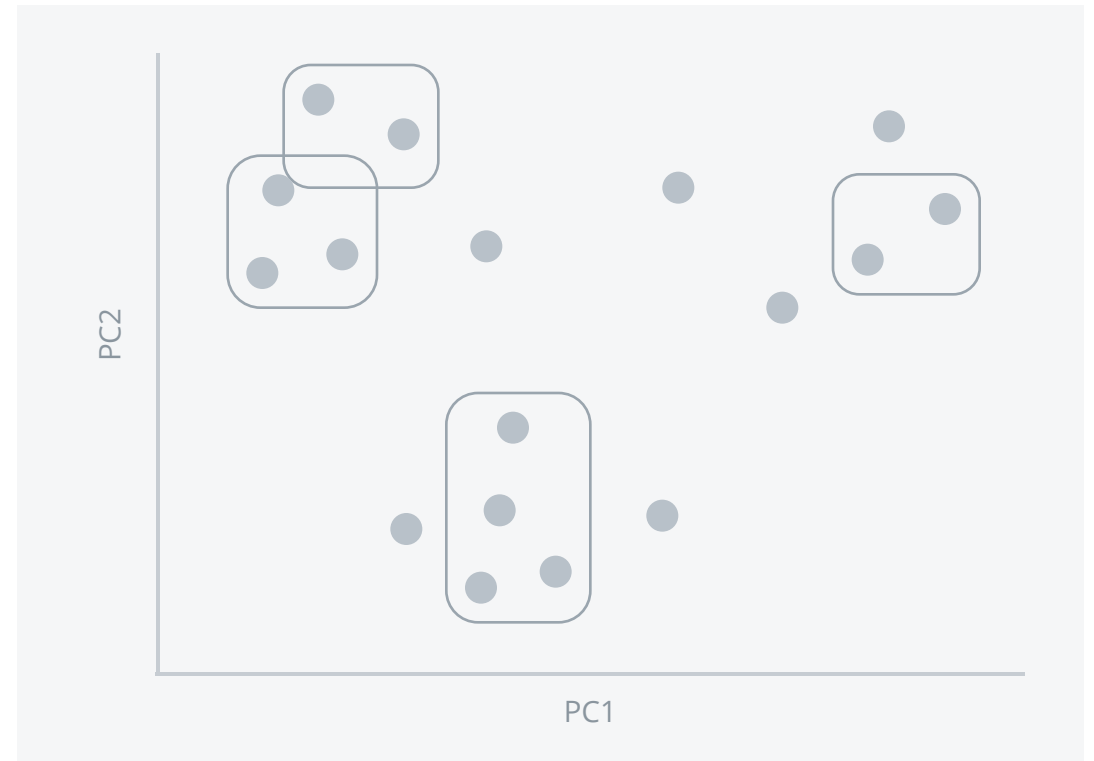


5 closest pairs merged · 5 pairs + 7 single = 12 clusters

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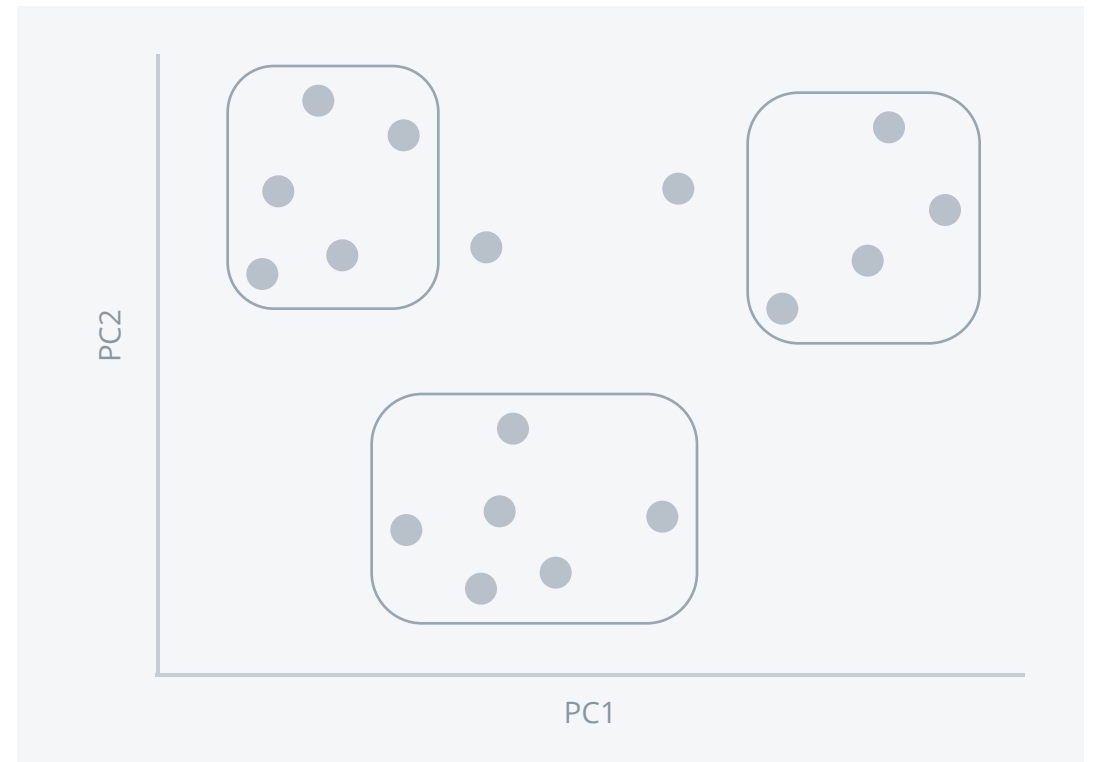


7 pairwise merges · 4 boxed + 6 single = 10 clusters

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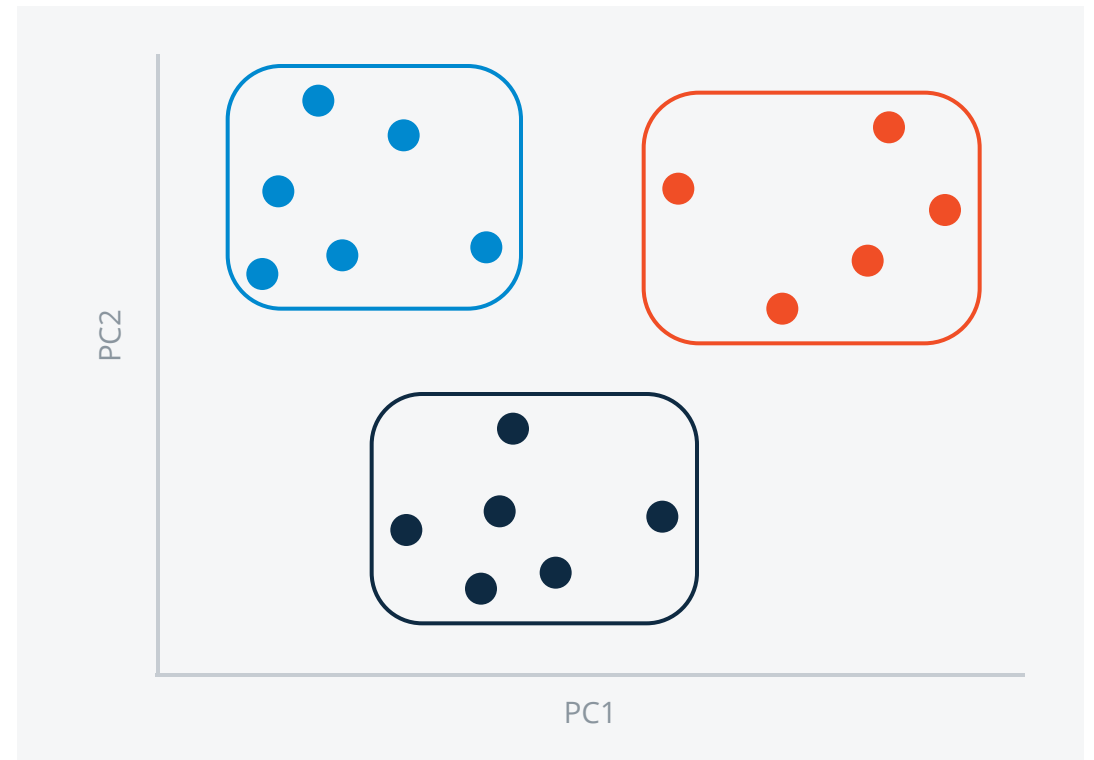


merging continues · 5 clusters left

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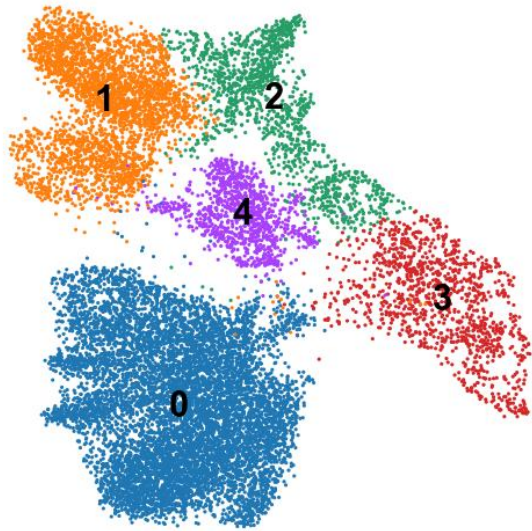
cut before the 59.5 $\rightarrow$ 68.7 jump $\cdot$ 6 / 5 / 6 cells

PART TWO

Choosing how finely to cut

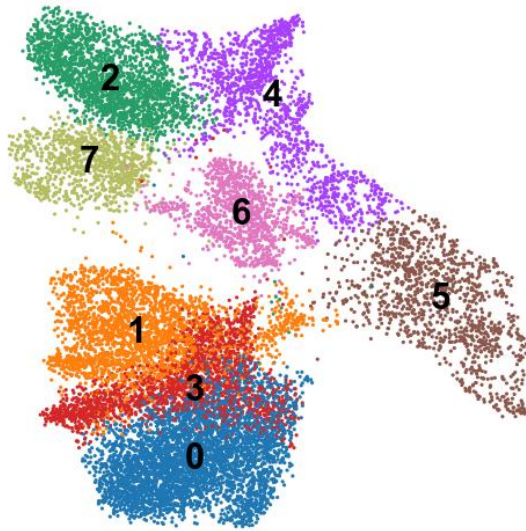
THE ONE PARAMETER YOU WILL TURN

Resolution decides how finely the same graph is cut



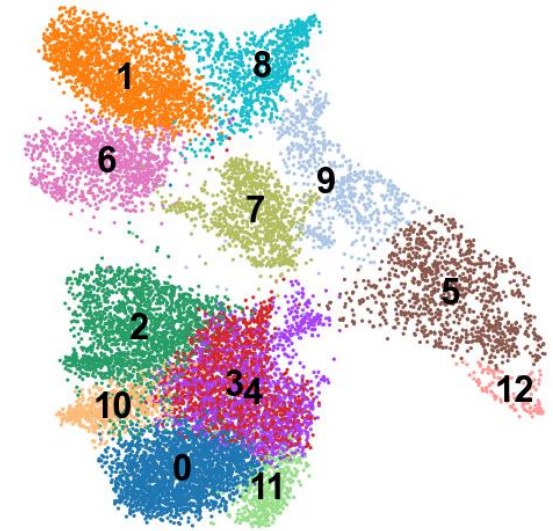
resolution 0.25 · 5 clusters

Broad lineages. Related subtypes stay merged.



resolution 0.5 · 8 clusters

Cell types. The usual starting point.



resolution 1.0 · 13 clusters

States, subsets — and noise split into groups.

There is no correct number of clusters — only a number that answers your question

Broad immune compartments or more detailed CD8 states?
Both are valid views of the same graph, at different resolutions.

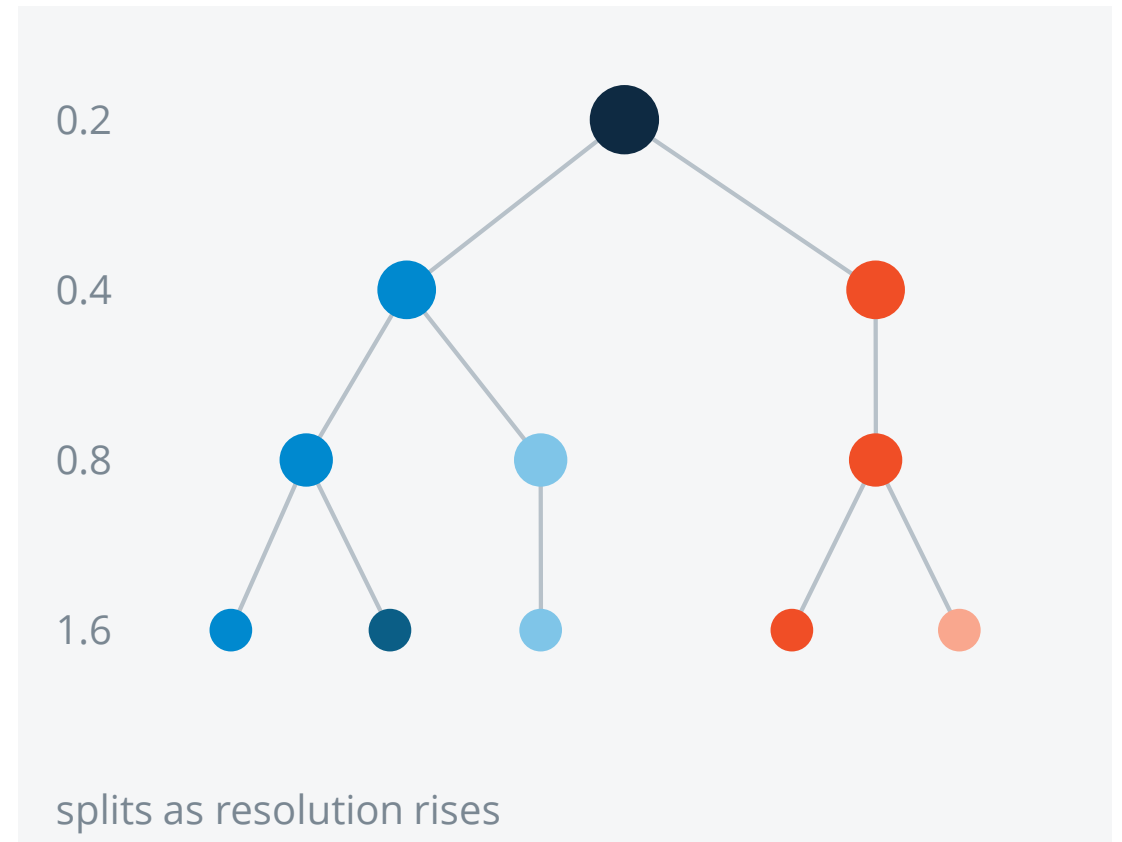
PRACTICE

Run a range of resolutions instead of trusting a single one

Cluster at 0.2, 0.4, 0.8, 1.6 and follow what happens to each group as the resolution rises.

Clusters that split cleanly and stay split are real structure. Clusters whose cells keep swapping between groups are the algorithm cutting noise.

Then check the markers. If a split produces no genes you can defend, it is not a split worth keeping.



FAILURE MODES

Under- and over-clustering fail in opposite directions

Too few clusters

Distinct populations are merged. Marker lists mix two identities, and rare cell types disappear entirely.

Symptom: a cluster with markers of two lineages, or visible sub-structure inside one cluster on the UMAP.

Too many clusters

Noise, cell-cycle phase or ambient RNA get their own clusters. Every cluster is significant against every other, because the genes that made the split are the genes you then test — "double dipping".

Symptom: neighbouring clusters with no distinguishing genes beyond ribosomal or mitochondrial content.

SANITY CHECKS

A cluster you can trust survives small changes to the analysis

Resampling

Re-cluster bootstrapped subsets of the cells; stable clusters reappear almost unchanged.

Parameter jitter

Change k , the number of PCs, or the random seed. Real populations tolerate it.

Silhouette width

How much closer a cell sits to its own cluster than to the next one. Low values mean a soft, arbitrary boundary.

Biological reading

The decisive test: does the cluster have markers a biologist recognises and can name?

WATCH OUT

Most clustering problems come from a handful of habits

- 01** Clustering on UMAP coordinates instead of the PCA or integrated embedding.
- 02** Reporting the default resolution as if it were a result rather than a setting.
- 03** Clustering unintegrated data, then annotating clusters that are really samples.
- 04** Forgetting that doublets and dying cells form their own convincing clusters.
- 05** Not recording k , the number of PCs, the resolution and the seed. Results become irreproducible.

CHEAT SHEET

Clustering, in the order you will actually do it

- 1 Choose the embedding: PCA, or the integrated latent space if you have batches.
- 2 Build the KNN / SNN graph with $k \approx 10\text{--}30$ neighbours.
- 3 Run Leiden at several resolutions ($0.2 \rightarrow 2.0$), not one.
- 4 Colour the UMAP by cluster, by sample and by QC metrics — check nothing tracks a technical variable.
- 5 Compute markers per cluster and name what you can.
- 6 Merge clusters without defensible markers; split further only where the biology asks.
- 7 Record every parameter and the random seed.
- 8 Report the resolution you chose and why, alongside the figure.

FURTHER READING

Where to go next

Single-cell best practices — cellular structure & clustering · sc-best-practices.org

Orchestrating Single-Cell Analysis with Bioconductor — clustering chapter · bioconductor.org/books/OSCA

Seurat guided clustering tutorial · satijalab.org/seurat

Analysis of single cell RNA-seq data — biological analysis · singlecellcourse.org

Traag, Waltman & van Eck (2019), *From Louvain to Leiden* · Scientific Reports

Thank you

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